

against software attacks because of the viruses' ability to change permission sets with each new owner that is infected. A virus attack gains access to multiple users through shared files. Access control is designed to allow the sharing of files. The ability to share files is a basic need of the user and cannot be eliminated without destroying the usefulness of the system. Discretionary Access Control (DAC) is not protection against software attacks because it is a weak form of protection that can be bypassed and, as discretionary, is at the control of the end users who very often ignore it. Sites where the majority of the files on the system have no DAC protection are normal. (Many Unix sites have permission bit settings of 777 which allow anyone to read, write, execute or modify the file.) Mandatory Access Controls (MAC) also has little effect on virus activity for the same reasons, although MAC can be configured to be neither weak nor easy to bypass. Each time a virus attacks an executable file owned by a different user, it takes on the full privileges of that user, including access to files of other users whose permissions intersect the DAC and MAC permission sets of the infected user. On all systems, the need to share files forces the creation of users who exist in multiple permission sets. This multiple membership allows viruses to move between MAC compartments and levels. The reduction of multiple membership users will slow the advance of a virus but will not eliminate it. Finally, once a virus gains access to an operator account (root, operator, isso) it cannot be stopped by any form of control.

Inspection is the traditional way of locating both known holes in operating systems and in locating known viruses. The key word here is "known". System audit tools such as COPS, SATAN and others can only locate holes that are known to them. Virus scanners can only locate viruses that are known to them. This means that a virus scanner or inspection tool is obsolete even before it is shipped from the factory. It can only deal with the past, never the present or future since conditions searched for must exist at the time of coding. Virus scanner have to be constantly updated. This is becoming a problem with